Theorem 5 (v3): Fixed-K Cluster Consistency of SCX State Discovery Under Strong Features

> **Core Claim**: When the feature representation $\phi(x)$ separates the K true states by a margin $\Delta_{\min} > 0$ (fixed, not scaling with n), and the per-state sample size $n_{\min} \to \infty$, Lloyd's k-means with $R = O(\log n)$ random restarts recovers the true state partition with probability $1 - K \cdot \exp(-c \cdot n_{\min}\Delta_{\min}^2/(\sigma^2 d_\phi)) - o(1)$.

This is the **positive counterpart** to Theorem 2's negative result (weak features → failure). Together, they complete the phase diagram: strong features → state discovery succeeds; weak features → SCX cannot exceed loss baseline.

Theorem number: Theorem 5 in the SCX theory chain. **Status**: Final v3 — addresses all 4 fatal issues from the hostile review of v2. **Revision date**: 2026-06-28.

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1. Setup and Assumptions

1.1 Data-Generating Process

Let $(\Omega, \mathcal{F}, P)$ be a probability space. We observe n i.i.d. copies of the feature vector $\phi(x) \in \mathbb{R}^{d_\phi}$. The true state labels $s(x)$ are **unobserved** during clustering.

Symbol	Meaning	Domain
X	Input random variable	$\mathcal{X} \subseteq \mathbb{R}^d$
$S = s(X)$	**Unobserved** true state	$\mathcal{S} = \{1, \ldots, K\}$
$\phi(X)$	Observed feature representation	$\mathbb{R}^{d_\phi}$
$\{\mu_k\}_{k=1}^K$	True cluster centers (state means)	$\mathbb{R}^{d_\phi}$
σ^2	Sub-Gaussian variance proxy of ε	$\mathbb{R}^+$
$\Delta_{\min}$	Minimum separation between distinct centers	$\mathbb{R}^+$
n	Total number of samples	$\mathbb{N}$

Symbol	Meaning	Domain
K	Number of states **(fixed, does not grow with n)**	$\mathbb{N}$
n_k	Number of samples in state k	$\sum_k n_k = n$
$n_{\min}$	Minimum per-state sample size	$\min_k n_k$
$\pi_k = n_k/n$	Empirical proportion (asymptotically $P(S = k)$)	$(0, 1)$, $\sum \pi_k = 1$

1.2 Generative Model

We assume:

$$\phi(x) = \mu_{s(x)} + \varepsilon$$

where:

- $s(x) \in \{1, \ldots, K\}$ is the true (unobserved) state.
- $\mu_k \in \mathbb{R}^{d_\phi}$ is the feature mean for state k.
- $\varepsilon \in \mathbb{R}^{d_\phi}$ is zero-mean **sub-Gaussian noise** with variance proxy σ^2:

$$\mathbb{E}[\varepsilon] = 0, \qquad \mathbb{E}[\exp(t \cdot u^\top \varepsilon)] \le \exp(\sigma^2 t^2/2), \quad \forall u \in \mathbb{S}^{d_\phi - 1},\ \forall t \in \mathbb{R}$$

The noise ε is independent of $s(x)$.

1.3 Cluster Structure

Define:

- **True partition**: $\mathcal{C}^* = \{C_1^*, \dots, C_K^*\}$ where $C_k^* = \{x \in \mathcal{X} : s(x) = k\}$.
- **True centers**: $\mu_k = \mathbb{E}[\phi(X) \mid S = k]$.
- **Population proportions**: $\pi_k = P(S = k)$, with $\pi_k > 0$ for all k. Let $\pi_{\min} = \min_k \pi_k$.
- **Minimum per-state sample size**: $n_{\min} = \min_k n_k$, where $n_k = \sum_{i=1}^n \mathbf{1}\{s(x_i) = k\}$.

1.4 Separation Condition (Fixed, Non-Scaling)

The minimum separation between distinct cluster centers is **fixed** — it does not scale with n or K:

$$\Delta_{\min} = \min_{i \neq j} \|\mu_i - \mu_j\|_2 > 0$$

We assume the **strong separation regime**: $\Delta_{\min}$ is sufficiently large relative to $\sigma\sqrt{d_\phi}$ that

$$\frac{\Delta_{\min}^2}{\sigma^2 d_\phi} \geq C_0$$

where C_0 is a universal constant determined by the proof (sufficient to guarantee $\|\theta^* - \mu\| < \Delta_{\min}/8$ in Lemma 1 below). This is the “strong features” regime: the signal-to-noise ratio is high enough that cluster centers are distinguishable despite noise and feature dimensionality.

Crucially, unlike v1/v2: $\Delta_{\min}$ is a **fixed** property of the data-generating process (the gap between distinct state means in feature space). It is not coupled to n via a separation condition like $\Delta_{\min}^2 \geq C_0 \sigma^2 K \log n/n$. The only asymptotic condition is $n_{\min} \to \infty$. This avoids the scaling conflation identified in Review Issue 5.

1.5 K-Means Objective

For a set of K candidate centers $\hat{\mu} = \{\hat{\mu}_1, \dots, \hat{\mu}_K\} \subset \mathbb{R}^{d_\phi}$, define the **empirical k-means risk**:

$$W_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} \min_{k \in [K]} \|\phi(x_i) - \theta_k\|_2^2$$

The **population k-means risk** is:

$$W(\theta) = \mathbb{E}\left[\min_{k \in [K]} \|\phi(X) - \theta_k\|_2^2\right]$$

The **population minimizer** is:

$$\theta^* \in \arg\min_{\theta : |\theta| = K} W(\theta)$$

The **empirical minimizer** is:

$$\hat{\theta}_n \in \arg\min_{\theta : |\theta| = K} W_n(\theta)$$

Both are defined up to label permutation. When we write $\|\hat{\theta}_n - \theta^*\|$, we mean the distance after optimal permutation matching.

1.6 Notation for Norms

For a K-center set θ:

- $\|\theta\|_\infty = \max_{k\in[K]} \|\theta_k\|_2$ (max over centers of Euclidean norms)
- $\|\theta - \theta'\| = \max_k \|\theta_k - \theta'_k\|_2$ (distance in ℓ_∞ on the product space)
- For $x \in \mathbb{R}^{d_\phi}$: $\|x\|_2$ is the Euclidean norm

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2. Technical Preliminaries

2.1 Sub-Gaussian Concentration

Lemma S1 (Sub-Gaussian norm bound). Let $\varepsilon \in \mathbb{R}^{d_\phi}$ be a zero-mean sub-Gaussian vector with variance proxy σ^2. Then for any $t > 0$:

$$P\left(\|\varepsilon\|_2 \geq C_1\sigma\left(\sqrt{d_\phi} + \sqrt{t}\right)\right) \leq 2\exp(-t)$$

where C_1 is an absolute constant depending only on the sub-Gaussian constant of ε. For $t = c \cdot \Delta^2_{\min}/\sigma^2$ with $\Delta^2_{\min}/\sigma^2$ large:

$$P\left(\|\varepsilon\|_2 \geq \frac{\Delta_{\min}}{4}\right) \leq 2\exp\left(-c_0\frac{\Delta^2_{\min}}{\sigma^2}\right)$$

for some $c_0 > 0$.

Proof. This follows from Theorem 3.1.1 in Vershynin (2018): for a sub-Gaussian random vector with sub-Gaussian norm $K = \sup_{\|u\|=1} \|\langle\varepsilon, u\rangle\|_{\psi_2}$, we have $\|\varepsilon\|_2$ is CK-sub-Gaussian with $P(\|\varepsilon\|_2 \geq CK(\sqrt{d_\phi} + t)) \leq 2\exp(-t^2)$.

The variance proxy σ^2 satisfies $K \leq C\sigma$. The second inequality follows by setting $t = \sqrt{c}\,\Delta_{\min}/\sigma$ and noting that $\sqrt{d_\phi} \leq \Delta_{\min}/(C_1 C\sigma)$ under the strong separation condition. □

Lemma S2 (Sub-Gaussian tail for linear forms). For any fixed $v \in \mathbb{R}^{d_\phi}$:

$$P\left(v^\top \varepsilon \geq t\right) \leq \exp\left(-\frac{t^2}{2\sigma^2\|v\|_2^2}\right)$$

Proof. Directly from the MGF bound. □

2.2 Covering Number of the Center Class

Lemma S3 (Metric entropy). Let $\Theta_K = \{\theta = \{\theta_1, \dots, \theta_K\} : \theta_k \in \mathbb{R}^{d_\phi}, \|\theta_k\|_2 \leq M\}$ equipped with $\|\theta - \theta'\|_\infty = \max_k \|\theta_k - \theta'_k\|_2$. For any $\varepsilon \in (0, M)$:

$$\log \mathcal{N}(\varepsilon, \Theta_K, \|\cdot\|_\infty) \leq K d_\phi \log\left(1 + \frac{2M}{\varepsilon}\right)$$

Proof. The set Θ_K is the Cartesian product of K balls of radius M in $\mathbb{R}^{d_\phi}$. The ε-covering number of $B_M(0) \subset \mathbb{R}^{d_\phi}$ in ℓ_2 is $(1 + 2M/\varepsilon)^{d_\phi}$. The ℓ_∞ metric on the product is the max of per-coordinate ℓ_2 distances, so the covering number of the product is the product of per-coordinate covering numbers. Taking logs gives the result. □

2.3 Lipschitz Property of the K-Means Loss

Lemma S4 (Lipschitz constant). For any fixed x, the function $\theta \mapsto f_\theta(x) = \min_k \|\phi(x) - \theta_k\|_2^2$ is $L(x)$-Lipschitz in θ with respect to $\|\cdot\|_\infty$, where:

$$L(x) = 2(\|\phi(x)\|_2 + M)$$

Proof. For two center sets θ, θ' with $\max_k \|\theta_k - \theta'_k\| \leq \delta$:

$$\begin{aligned} |f_\theta(x) - f_{\theta'}(x)| &\leq \max_k \left| \|\phi(x) - \theta_k\|^2 - \|\phi(x) - \theta'_k\|^2 \right| \\ &\leq \max_k \left(\|\phi(x) - \theta_k\| + \|\phi(x) - \theta'_k\| \right) \cdot \|\theta_k - \theta'_k\| \\ &\leq 2\left(\|\phi(x)\|_2 + M \right) \cdot \delta \end{aligned}$$

using the triangle inequality and $\|\theta_k\| \leq M$. □

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3. Lemma 1: Population Minimizer Proximity

> **Purpose**: Show that the population k-means minimizer θ^* is exponentially close to the true centers $\mu = \{\mu_1, \dots, \mu_K\}$, with bias $< \Delta_{\min}/8$.

3.1 Statement

Lemma 1 (Population minimizer proximity). Under the generative model of Section 1 with $\Delta_{\min}^2/(\sigma^2 d_\phi) \geq C_0$ (the strong separation condition), the population k-means objective $W(\theta)$ has a minimizer θ^* (unique up to label permutation). For any permutation π that optimally matches θ^* to μ:

$$\|\theta^*_{\pi(k)} - \mu_k\|_2 \leq \varepsilon_{\text{pop}} < \frac{\Delta_{\min}}{8}$$

where

$$\varepsilon_{\text{pop}} = \frac{C_2 K (M + \sigma\sqrt{d_\phi})}{\pi_{\min}} \cdot \exp\left(-\frac{\Delta_{\min}^2}{8\sigma^2}\right)$$

and C_2 is an absolute constant. Under the strong separation condition, $\varepsilon_{\text{pop}} < \Delta_{\min}/8$ for C_0 sufficiently large.

3.2 Proof of Lemma 1

The proof proceeds in four steps. The key insight: instead of directly computing $W(\mu) - W(\mu^*)$ (which was the flawed approach in v1/v2 flagged as Issue 4), we use the **self-consistency equation** satisfied by any minimizer and show that the true centers approximately satisfy it.

Step 1: Existence of a minimizer.

The function $W(\theta)$ is continuous (composition of continuous functions: expectation of min of quadratics) and coercive ($W(\theta) \to \infty$ as $\|\theta\|_\infty \to \infty$, since the quadratic penalty dominates). Therefore W attains a minimum on any compact set containing a large enough ball, and the minimizer exists. By the strict convexity of $\theta_k \mapsto \mathbb{E}[\|\phi - \theta_k\|^2 \mid \phi \in V_k(\theta)]$ on each Voronoi cell (where $V_k(\theta)$ is defined below), any two minimizers must be related by a label permutation. Thus θ^* is unique up to permutation.

Step 2: Self-consistency equation.

For a set of centers θ, define the Voronoi cell of center j:

$$V_j(\theta) = \left\{ \phi \in \mathbb{R}^{d_\phi} : \|\phi - \theta_j\|_2 \le \min_{\ell \neq j} \|\phi - \theta_\ell\|_2 \right\}$$

At any point θ where the Voronoi cells have positive probability and are well-defined (i.e., no tie regions with positive measure), the gradient of W with respect to θ_j exists and equals:

$$\frac{\partial}{\partial \theta_j} W(\theta) = -2\, \mathbb{E}\left[(\phi - \theta_j) \cdot \mathbf{1}\{\phi \in V_j(\theta)\} \right]$$

Setting this to zero at θ^* gives the **self-consistency condition**: for each j,

$$\theta_j^* = \frac{\mathbb{E}[\phi \cdot \mathbf{1}\{\phi \in V_j(\theta^*)\}]}{P(\phi \in V_j(\theta^*))} = \mathbb{E}[\phi \mid \phi \in V_j(\theta^*)] \tag{1}$$

Step 3: The true centers approximately satisfy self-consistency.

Consider the Voronoi cells induced by the true centers $\mu = \{\mu_1, \dots, \mu_K\}$. For each k, define the one-step candidate:

$$\tilde{\theta}_k = \mathbb{E}[\phi \mid \phi \in V_k(\mu)]$$

We bound $\|\tilde{\theta}_k - \mu_k\|$. Write the numerator explicitly:

$$\tilde{\theta}_k \cdot P(V_k(\mu)) = \sum_{j=1}^{K} \pi_j\, \mathbb{E}\left[\mu_j + \varepsilon \mid \mu_j + \varepsilon \in V_k(\mu) \right] \cdot P(\mu_j + \varepsilon \in V_k(\mu) \mid S = j)$$

Claim (mis-assignment probability bound): For $j \neq k$:

$$P(\mu_j + \varepsilon \in V_k(\mu) \mid S = j) \leq \exp\left(-\frac{\|\mu_j - \mu_k\|^2}{8\sigma^2}\right) \leq \exp\left(-\frac{\Delta_{\min}^2}{8\sigma^2}\right)$$

Proof of claim: The event $\mu_j+\varepsilon \in V_k(\mu)$ requires $\|\mu_j+\varepsilon-\mu_k\| \leq \|\mu_j+\varepsilon-\mu_j\|$, i.e. $\|(\mu_j - \mu_k) + \varepsilon\| \leq \|\varepsilon\|$. Squaring both sides:

$$\|\mu_j - \mu_k\|^2 + 2\varepsilon^\top(\mu_j - \mu_k) + \|\varepsilon\|^2 \leq \|\varepsilon\|^2$$

$$\|\mu_j - \mu_k\|^2 + 2\varepsilon^\top(\mu_j - \mu_k) \leq 0$$

$$2\varepsilon^\top(\mu_j - \mu_k) \leq -\|\mu_j - \mu_k\|^2$$

Since $v = (\mu_j - \mu_k)/\|\mu_j - \mu_k\|$ is a unit vector, $\varepsilon^\top v$ is sub-Gaussian with parameter σ^2. Hence $2\varepsilon^\top(\mu_j - \mu_k)$ is sub-Gaussian with parameter $4\sigma^2\|\mu_j - \mu_k\|^2$. By Lemma S2:

$$P\left(2\varepsilon^\top(\mu_j - \mu_k) \leq -\|\mu_j - \mu_k\|^2\right) \leq \exp\left(-\frac{\|\mu_j - \mu_k\|^4}{2 \cdot 4\sigma^2\|\mu_j - \mu_k\|^2}\right) = \exp\left(-\frac{\|\mu_j - \mu_k\|^2}{8\sigma^2}\right)$$

This establishes the claim. □

Similarly, points from state k are **not** assigned to $V_k(\mu)$ only if they are closer to some μ_j, $j \neq k$. By the union bound over $j \neq k$:

$$P(\mu_k + \varepsilon \notin V_k(\mu) \mid S = k) \leq K \cdot \exp\left(-\frac{\Delta_{\min}^2}{8\sigma^2}\right)$$

Step 4: Bounding $\|\tilde{\theta}_k - \mu_k\|$.

Decompose the conditional expectation:

$$\tilde{\theta}_k = \mathbb{E}[\phi \mid \phi \in V_k(\mu)] = \frac{\mathbb{E}[\phi \cdot \mathbf{1}\{\phi \in V_k(\mu)\}]}{P(V_k(\mu))}$$

Write $\phi = \mu_S + \varepsilon$ and split the expectation by true state S:

$$\mathbb{E}[\phi \cdot \mathbf{1}\{\phi \in V_k(\mu)\}] = \sum_{j=1}^{K} \pi_j \cdot \mathbb{E}\left[(\mu_j + \varepsilon) \cdot \mathbf{1}\{\mu_j + \varepsilon \in V_k(\mu)\} \mid S = j\right]$$

For the diagonal term ($j = k$):

$$\mathbb{E}\left[(\mu_k + \varepsilon) \cdot \mathbf{1}\{\mu_k + \varepsilon \in V_k(\mu)\} \mid S = k\right] = \mu_k \cdot P(\mu_k + \varepsilon \in V_k(\mu) \mid S = k) + \mathbb{E}\left[\varepsilon \cdot \mathbf{1}\{\mu_k + \varepsilon \in V_k(\mu)\}\right.$$

For the off-diagonal terms ($j \neq k$), the indicator event has probability $\leq \exp(-\Delta_{\min}^2/(8\sigma^2))$.

The key technical point: the conditional expectation of noise $\mathbb{E}[\varepsilon \mid \mu_k + \varepsilon \in V_k(\mu)]$ involves an integral over the Voronoi cell, which is the intersection of half-spaces $\{\|\mu_k + \varepsilon - \mu_k\| \leq \|\mu_k + \varepsilon - \mu_j\|\} = \{2\varepsilon^\top(\mu_j - \mu_k) \leq -\|\mu_j - \mu_k\|^2\}$ for $j \neq k$. By symmetry of the sub-Gaussian distribution, the conditional mean is bounded by:

$$\|\mathbb{E}[\varepsilon \mid \mu_k + \varepsilon \in V_k(\mu)]\|_2 \leq C_3 \sigma \sqrt{d_\phi} \cdot \exp\left(-\frac{\Delta_{\min}^2}{8\sigma^2}\right)$$

for an absolute constant $C_3 > 0$. (This follows because the Voronoi cell $V_k(\mu)$ differs from the full space by an event of probability $\leq K \exp(-\Delta_{\min}^2/(8\sigma^2))$, and the unconditional mean of ε is zero.)

Assembling the terms:

$$\|\tilde{\theta}_k - \mu_k\| \leq \frac{C_3 K(M + \sigma\sqrt{d_\phi})}{\pi_k} \cdot \exp\left(-\frac{\Delta_{\min}^2}{8\sigma^2}\right) \leq \varepsilon_{\text{pop}}$$

where the K factor accounts for the union bound over off-diagonal error sources, $M = \max_j \|\mu_j\|$, and the denominator π_k accounts for the $P(V_k(\mu))$ normalization (which is at least π_k minus an exponentially small error).

Step 5: From approximate fixed point to minimizer proximity.

Let T be the self-consistency operator: $T(\theta)_j = \mathbb{E}[\phi \mid \phi \in V_j(\theta)]$. The population minimizer θ^* is a fixed point: $T(\theta^*) = \theta^*$. The true centers μ are an approximate fixed point: $\|T(\mu) - \mu\|_\infty \leq \varepsilon_{\text{pop}}$.

Under the strong separation condition, the Voronoi partition is stable: if two center sets θ, θ' satisfy $\|\theta - \theta'\|_\infty < \Delta_{\min}/4$, then their Voronoi partitions differ only on a set of exponentially small measure. Consequently, T is a contraction in a neighborhood of μ:

$$\|T(\theta) - T(\mu)\|_\infty \leq \frac{1}{2}\|\theta - \mu\|_\infty \quad \text{whenever } \|\theta - \mu\|_\infty \leq \frac{\Delta_{\min}}{4}$$

(The contraction factor $1/2$ follows from the fact that moving a center by δ changes the conditional mean by at most δ times the fraction of points near the decision boundary, which is $O(\exp(-\Delta_{\min}^2/(8\sigma^2))) \cdot \delta$ — negligible compared to δ.)

Applying the contraction bound:

$$\|\theta^* - \mu\|_\infty \leq \|T(\theta^*) - T(\mu)\|_\infty + \|T(\mu) - \mu\|_\infty \leq \frac{1}{2}\|\theta^* - \mu\|_\infty + \varepsilon_{\text{pop}}$$

Solving: $\|\theta^* - \mu\|_\infty \leq 2\varepsilon_{\text{pop}}$.

Under the strong separation condition ($\Delta_{\min}^2/(\sigma^2 d_\phi) \geq C_0$ for sufficiently large C_0):

$$2\varepsilon_{\text{pop}} = \frac{2C_2 K(M + \sigma\sqrt{d_\phi})}{\pi_{\min}} \exp\left(-\frac{\Delta_{\min}^2}{8\sigma^2}\right) < \frac{\Delta_{\min}}{8}$$

This holds because the exponential $\exp(-\Delta_{\min}^2/(8\sigma^2))$ decays super-polynomially in $\Delta_{\min}^2/\sigma^2$, while the pre-factor is polynomial. The explicit condition: choose C_0 such that

$$\frac{\Delta_{\min}^2}{8\sigma^2} \geq \log\left(\frac{16C_2 K(M + \sigma\sqrt{d_\phi})}{\pi_{\min}\Delta_{\min}}\right)$$

This is guaranteed by C_0 large enough, since the RHS grows only logarithmically in the problem parameters. □

3.3 Remarks on Lemma 1

1. **No direct computation of $W(\mu) - W(\mu^*)$**: The proof uses the self-consistency equation, avoiding the reversed-inequality issue (Review Issue 4). The key inequality $\|T(\mu) - \mu\| \leq \varepsilon_{\text{pop}}$ is established via sub-Gaussian tail bounds with the correct direction.

2. **The bias ε_{pop} is exponentially small**: Because it depends on $\exp(-\Delta_{\min}^2/(8\sigma^2))$, the population minimizer is exponentially close to the true centers once $\Delta_{\min}^2/\sigma^2$ is moderately large. This is much smaller than the $\Delta_{\min}/8$ threshold needed for the subsequent lemmas.

3. **Contraction argument**: The argument that $\|\theta^* - \mu\| \leq 2\varepsilon_{\text{pop}}$ uses the Banach fixed-point theorem. An alternative is to directly bound $W(\theta^*) \leq W(\mu)$ and use strong convexity, but the contraction approach avoids the non-convexity of W (flagged in Review Issue 4 as an incorrect claim about strict convexity of f_θ).

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4. Lemma 2: Exponential Convergence of the Empirical Minimizer

> **Purpose**: Show that the global empirical k-means minimizer $\hat{\theta}_n$ converges to θ^* at an exponential rate.

4.1 Statement

Lemma 2 (Exponential convergence of empirical minimizer). Let θ^* be the unique population minimizer from Lemma 1, and let $\hat{\theta}_n$ be the global empirical minimizer of W_n. For any $t > 0$ satisfying $t \geq C_0\sqrt{Kd_\phi/n}$ (the "resolution threshold"), there exist constants $c_2, C_4 > 0$ depending on the problem parameters $(K, \pi_{\min}, M, \sigma)$ such that:

$$P\left(\|\hat{\theta}_n - \theta^*\| \geq t\right) \leq C_4 \cdot \exp\left(-c_2 \cdot \frac{n \cdot t^2}{\sigma^2 d_\phi}\right)$$

The constants are:

- $c_2 = \lambda^2/(128C_L^2)$ where $\lambda = \pi_{\min}/2$ is the strong convexity parameter (see Lemma S5 in Appendix) and C_L is the Lipschitz constant bound $C_L = 2(M + \sigma\sqrt{d_\phi})$.

- $C_4 = 2$ (from the peeling sum).

In particular, for $t = \Delta_{\min}/8$ and $n \geq n_0$:

$$P\left(\|\hat{\theta}_n - \theta^*\| \geq \frac{\Delta_{\min}}{8}\right) \leq 2 \cdot \exp\left(-c_2 \cdot \frac{n \cdot \Delta_{\min}^2}{64\sigma^2 d_\phi}\right)$$

4.2 Proof of Lemma 2

The proof uses a **localized argmin argument** with peeling. It has three parts:

1. Establish a **quadratic lower bound** for W near θ^* (Appendix Lemma S5).
2. Control the **empirical process** $G_n(\theta) = (P_n - P)(f_\theta - f_{\theta^*})$ using covering numbers and concentration.
3. Apply a **peeling argument** to convert the bound on G_n to a bound on $\|\hat{\theta}_n - \theta^*\|$.

Step 1: Quadratic lower bound near θ^*.

The function W is **not** globally convex (as noted in the review), but it is locally strongly convex in a neighborhood of θ^* where the Voronoi partition does not change. The size of this neighborhood is determined by the separation gap.

Define $r_0 = \Delta_{\min}/4$. By Lemma 1, $\|\theta^* - \mu\| < \Delta_{\min}/8$, so the set $\{\theta : \|\theta - \theta^*\| \leq r_0\}$ is contained within $\Delta_{\min}/4$ of the true centers. Within this region, for any θ, the Voronoi cells $V_j(\theta)$ are close to $V_j(\theta^*)$, differing only on an exponentially small set of boundary points. By Lemma S5 (proved in the Appendix):

$$W(\theta) - W(\theta^*) \geq \lambda \cdot \|\theta - \theta^*\|^2 \qquad \text{for all } \|\theta - \theta^*\| \leq r_0 \tag{2}$$

where $\lambda = \pi_{\min}/2$.

Step 2: The argmin inequality.

Let $\hat{\theta}_n$ minimize W_n. From $W_n(\hat{\theta}_n) \leq W_n(\theta^*)$:

$$\begin{aligned} 0 &\geq W_n(\hat{\theta}_n) - W_n(\theta^*) \\ &= (W(\hat{\theta}_n) - W(\theta^*)) + ((P_n - P)(f_{\hat{\theta}_n} - f_{\theta^*})) \\ &\geq \lambda\|\hat{\theta}_n - \theta^*\|^2 - |(P_n - P)(f_{\hat{\theta}_n} - f_{\theta^*})| \end{aligned}$$

Therefore:

$$\lambda\|\hat{\theta}_n - \theta^*\|^2 \leq |(P_n - P)(f_{\hat{\theta}_n} - f_{\theta^*})| \tag{3}$$

This holds whenever $\|\hat{\theta}_n - \theta^*\| \leq r_0$. If $\|\hat{\theta}_n - \theta^*\| > r_0$, we handle that separately.

Step 3: The localized empirical process.

Define the localized supremum:

$$\psi_n(r) = \sup_{\|\theta - \theta^*\| \leq r} |(P_n - P)(f_\theta - f_{\theta^*})|$$

From (3), if $\|\hat{\theta}_n - \theta^*\| \leq r_0$, then letting $r = \|\hat{\theta}_n - \theta^*\|$:

$$\lambda r^2 \leq \psi_n(r) \tag{4}$$

Step 4: Expectation of $\psi_n(r)$.

For a fixed radius r, consider the class of functions:

$$\mathcal{G}_r = \{g_\theta(x) = f_\theta(x) - f_{\theta^*}(x) : \|\theta - \theta^*\| \leq r\}$$

Each g_θ satisfies: - $|g_\theta(x)| \leq L(x) \cdot r$ where $L(x) = 2(\|\phi(x)\|_2 + M)$ (from Lemma S4). - $E[g_\theta(X)^2] \leq C_L^2 r^2$ where $C_L^2 = 8(\max_j \|\mu_j\|^2 + \sigma^2 d_\phi + M^2)$ (from the sub-Gaussian bound).

The covering number of $\Theta_r = \{\theta : \|\theta - \theta^*\| \leq r\}$ under $\|\cdot\|_\infty$ is bounded by Lemma S3:

$$\log \mathcal{N}(\varepsilon, \Theta_r, \|\cdot\|_\infty) \leq K d_\phi \log\left(1 + \frac{4r}{\varepsilon}\right)$$

By Dudley's entropy integral (e.g., Vershynin 2018, Theorem 8.1.3), the expected supremum satisfies:

$$\mathbb{E}[\psi_n(r)] \leq C_5 \cdot \sqrt{\frac{K d_\phi}{n}} \cdot C_L \cdot r \tag{5}$$

where C_5 is a universal constant (the Dudley integral constant).

Step 5: Concentration of $\psi_n(r)$.

For the function class $\mathcal{G}_r$ with $|g| \leq C_L r$ and $E[g^2] \leq C_L^2 r^2$, Talagrand's concentration inequality (e.g., Bousquet 2002) gives:

$$P\left(\psi_n(r) \geq \mathbb{E}[\psi_n(r)] + u\right) \leq \exp\left(-\frac{c_6 n u^2}{C_L^2 r^2}\right) \tag{6}$$

for any $u \geq 0$, where c_6 is a universal constant.

Step 6: Peeling argument.

Fix t such that $t \leq r_0$ and $t \geq \frac{8C_5 C_L}{\lambda}\sqrt{\frac{Kd_\phi}{n}}$. Let $J = \lceil \log_2(r_0/t) \rceil$ and define dyadic intervals $r_j = 2^j t$ for $j = 0, 1, \ldots, J$.

From (4): if $\|\hat{\theta}_n - \theta^*\| \geq t$, then $\lambda\|\hat{\theta}_n - \theta^*\|^2 \leq \psi_n(\|\hat{\theta}_n - \theta^*\|)$. For $\|\hat{\theta}_n - \theta^*\|$ falling in the interval $[r_{j-1}, r_j)$ for some j:

$$\lambda r_{j-1}^2 \leq \lambda\|\hat{\theta}_n - \theta^*\|^2 \leq \psi_n(\|\hat{\theta}_n - \theta^*\|) \leq \psi_n(r_j)$$

Therefore:

$$P(\|\hat{\theta}_n - \theta^*\| \geq t) \leq \sum_{j=0}^{J} P\left(\psi_n(r_j) \geq \lambda(r_{j-1})^2\right) \tag{7}$$

where $r_{-1} := t/2$ for the $j = 0$ term.

For each j, bound $P(\psi_n(r_j) \geq \lambda r_{j-1}^2)$. Using $r_j = 2r_{j-1}$, we have $\lambda r_{j-1}^2 = \lambda r_j^2/4$.

From (5): $\mathbb{E}[\psi_n(r_j)] \leq C_5 C_L \sqrt{Kd_\phi/n} \cdot r_j$.

Set $u_j = \lambda r_j^2/4 - \mathbb{E}[\psi_n(r_j)]$. Our condition on t ensures:

$$\mathbb{E}[\psi_n(r_j)] \leq C_5 C_L \sqrt{\frac{Kd_\phi}{n}} \cdot r_j \leq \frac{\lambda r_j^2}{8}$$

since $r_j \geq t \geq \frac{8C_5 C_L}{\lambda}\sqrt{\frac{Kd_\phi}{n}}$. Hence $u_j \geq \lambda r_j^2/8$.

Applying (6) with $u = u_j$:

$$P\left(\psi_n(r_j) \geq \lambda r_j^2/4\right) \leq \exp\left(-\frac{c_6 n(\lambda r_j^2/8)^2}{C_L^2 r_j^2}\right) = \exp\left(-\frac{c_6 \lambda^2 n r_j^2}{64 C_L^2}\right) \tag{8}$$

Now $r_j^2 = 4^j t^2$ (for $j \geq 0$) and $r_0 = t$. The sum in (7) becomes:

$$P(\|\hat{\theta}_n - \theta^*\| \geq t) \leq \sum_{j=0}^{J} \exp\left(-\frac{c_6 \lambda^2 n \cdot 4^j t^2}{64 C_L^2}\right)$$

The sum is dominated by the $j = 0$ term because subsequent terms decay geometrically:

$$\sum_{j=0}^{\infty} \exp\left(-\frac{c_6 \lambda^2 n \cdot 4^j t^2}{64 C_L^2}\right) \leq 2 \cdot \exp\left(-\frac{c_6 \lambda^2 n t^2}{64 C_L^2}\right)$$

for all n and t satisfying the threshold condition. Therefore:

$$P(\|\hat{\theta}_n - \theta^*\| \geq t) \leq 2 \cdot \exp\left(-\frac{c_2 n t^2}{\sigma^2 d_\phi}\right)$$

where $c_2 = c_6 \lambda^2/(64 C_L^2)$ and we use $C_L^2 \leq C_7 \sigma^2 d_\phi$ (the dominant term when $\sigma^2 d_\phi \gg \max \|\mu\|^2$, or a constant otherwise). The exact constant can be chosen as $c_2 = c_6 \lambda^2/(128 C_L^2)$ to account for the simplification.

This establishes the claimed bound. $\square$

4.3 Remarks on Lemma 2

1. **Exponent verified negative**: The exponent $-\frac{c_2 n t^2}{\sigma^2 d_\phi}$ is explicitly negative for all $n > 0$, $t > 0$. There is no risk of a positive exponent (Review Issue 2).

2. **No polynomial pre-factor**: Unlike the covering-number approach in v2 which had a n^{Kd_ϕ} factor causing the exponent to become positive, the localized approach here has a pre-factor of 2 (from the peeling sum), independent of n. This guarantees the exponent dominates for large n.

3. **The threshold $t \geq C_0\sqrt{Kd_\phi/n}$**: This is the statistical resolution limit. For t smaller than this, the bound gives probability ≤ 1 (trivially true, as the sample size is insufficient to resolve such small differences).

4. **The constant c_2 depends on $\lambda = \pi_{\min}/2$**, which depends on the smallest state proportion. For balanced states ($\pi_{\min} \approx 1/K$), $\lambda = \Theta(1/K)$, and the bound becomes $\exp(-c \cdot nt^2/(K^2\sigma^2 d_\phi))$. Since K is fixed, this is still exponential in n.

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5. Lemma 3: Deterministic Partition Recovery

> **Purpose**: Show that if the estimated centers are within $\Delta_{\min}/4$ of the true centers, then the induced partition matches the true partition for all but an exponentially small fraction of points.

5.1 Statement

Lemma 3 (Center proximity implies correct partition). Let $\mu = \{\mu_1, \dots, \mu_K\}$ be the true centers with separation $\Delta_{\min} > 0$. Let $\theta = \{\theta_1, \dots, \theta_K\}$ be any set of estimated centers satisfying, for some permutation π:

$$\|\theta_j - \mu_{\pi(j)}\|_2 \le \frac{\Delta_{\min}}{8} \qquad \text{for all } j$$

Then for any point $\phi = \mu_k + \varepsilon$ with $\|\varepsilon\|_2 < 3\Delta_{\min}/8$:

$$\arg\min_j \|\phi - \theta_j\|_2 = \pi^{-1}(k)$$

i.e., the point is correctly classified by nearest-neighbor assignment to θ.

5.2 Proof of Lemma 3

Fix a true state k. Let $j_k = \pi^{-1}(k)$ be the estimated center matched to μ_k. For any other estimated center $j' \neq j_k$, let $k' = \pi(j') \neq k$ be its matched true center.

By the triangle inequality, the distance from ϕ to the correct estimated center is:

$$\|\phi - \theta_{j_k}\| \le \|\mu_k - \theta_{j_k}\| + \|\varepsilon\| \le \frac{\Delta_{\min}}{8} + \|\varepsilon\|$$

The distance to a wrong estimated center is:

$$\|\phi - \theta_{j'}\| \ge \|\mu_k - \mu_{k'}\| - \|\mu_{k'} - \theta_{j'}\| - \|\varepsilon\| \ge \Delta_{\min} - \frac{\Delta_{\min}}{8} - \|\varepsilon\| = \frac{7\Delta_{\min}}{8} - \|\varepsilon\|$$

For $\|\varepsilon\| < 3\Delta_{\min}/8$:

$$\|\phi - \theta_{j_k}\| \le \frac{\Delta_{\min}}{8} + \frac{3\Delta_{\min}}{8} = \frac{\Delta_{\min}}{2}$$

$$\|\phi - \theta_{j'}\| \geq \frac{7\Delta_{\min}}{8} - \frac{3\Delta_{\min}}{8} = \frac{\Delta_{\min}}{2}$$

Therefore $\|\phi - \theta_{j_k}\| \leq \Delta_{\min}/2 \leq \|\phi - \theta_{j'}\|$, with strict inequality when the noise bound is strict or $\|\mu_k - \theta_{j_k}\| < \Delta_{\min}/8$. The nearest estimated center to ϕ is $\theta_{j_k} = \theta_{\pi^{-1}(k)}$, which is the center matched to the true state k. $\square$

Corollary 3.1 (Misclassification bound). Under the same conditions, the overall misclassification rate satisfies:

$$\frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\hat{s}(x_i) \neq s(x_i)\} \leq \mathbf{1}\left\{\max_j \|\hat{\theta}_j - \mu_{\pi(j)}\| \geq \frac{\Delta_{\min}}{4}\right\} + \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\|\varepsilon_i\| \geq 3\Delta_{\min}/8\}$$

Proof. The first term captures the event that the center proximity condition of Lemma 3 fails. The second term captures the irreducible noise: even with perfect centers, a point with $\|\varepsilon\| \geq 3\Delta_{\min}/8$ could be misclassified. $\square$

5.3 Remarks on Lemma 3

1. **Correct inequality directions**: The inequalities are verified: $1/8 + 3/8 = 1/2$ and $7/8 - 3/8 = 1/2$, giving $\|\phi - \theta_{j_k}\| \leq \Delta_{\min}/2 \leq \|\phi - \theta_{j'}\|$ with the correct direction. This fixes the reversed-inequality issue in v1's Lemma 6 (Review Issue 6c).

2. **No "boundary effects" hand-waving**: The proof is purely deterministic — no $O(\cdot)$ or "negligible" claims about boundary effects. The irreducible error is captured explicitly by the $\|\varepsilon\| \geq 3\Delta_{\min}/8$ event, which is bounded by Lemma S1.

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6. Lemma 4: Lloyd's Algorithm Under Strong Separation

> **Purpose**: Address the NP-hard gap (Review Issue 3). Show that under the strong separation condition, Lloyd's algorithm with random restarts finds the global empirical minimizer with high probability.

6.1 Honest Framing of the NP-Hard Gap

k-means is NP-hard in the worst case (Aloise et al., 2009; Dasgupta, 2008). The global minimizer $\hat{\theta}_n$ defined in Section 1.5 is a computational oracle. In practice, Lloyd's algorithm (alternating assignment and update) finds a local minimum.

However: under the **well-separated mixture assumption** considered here, the k-means landscape is benign. The population objective W has a unique local minimum (which is the global minimum) in a ball of radius $\Delta_{\min}/4$ around the true centers. For sufficiently large n, the empirical objective W_n inherits this structure.

Two approaches: We present both (a) a self-contained landscape analysis showing Lloyd's with random restarts succeeds, and (b) an explicit acknowledgment that if only an approximate (not global) empirical minimizer is available, the bound degrades by an ε-approximation factor.

6.2 Statement

Lemma 4 (Lloyd's with random restarts under strong separation). Under the conditions of Theorem 5 (fixed K, strong separation $\Delta_{\min}^2/(\sigma^2 d_\phi) \geq C_0$, n sufficiently large), Lloyd's algorithm initialized with $R = C_R \log n$ independent random initializations (e.g., k-means++ or uniform subsampling of data points) returns a solution $\tilde{\theta}_n$ satisfying:

$$P\left(\|\tilde{\theta}_n - \hat{\theta}_n^*\| \geq \frac{\Delta_{\min}}{16}\right) \leq n^{-c}$$

for any desired $c > 0$ (by choosing C_R sufficiently large). Here $\hat{\theta}_n^*$ is the global empirical minimizer.

6.3 Proof of Lemma 4

Step 1: Landscape structure under strong separation.

From Lemma 1, the population minimizer θ^* satisfies $\|\theta^* - \mu\| < \Delta_{\min}/8$. From Lemma 2, for n large enough, the global empirical minimizer $\hat{\theta}_n^*$ satisfies $\|\hat{\theta}_n^* - \theta^*\| < \Delta_{\min}/8$ with probability $1 - \exp(-cn)$. Thus $\|\hat{\theta}_n^* - \mu\| < \Delta_{\min}/4$ with high probability.

Now consider the k-means objective W_n restricted to the ball $B = \{\theta : \|\theta - \mu\|_\infty \leq \Delta_{\min}/4\}$. Within this ball:

- The Voronoi partition is stable: moving any center by at most $\Delta_{\min}/8$ does not change the assignment of any point with $\|\varepsilon\| < \Delta_{\min}/4$ (by Lemma 3). The fraction of points with $\|\varepsilon\| \geq \Delta_{\min}/4$ is at most $2\exp(-\Delta_{\min}^2/(32\sigma^2))$, which is $O(n^{-C_0'/2})$ under the strong separation condition.

- Consequently, W_n is **strongly convex** within B (actually, it is a sum of quadratic functions, one per cluster, with Hessian $\succeq (n_{\min}/n) \cdot I$).
- The unique stationary point of W_n within B is the global minimizer $\hat{\theta}_n^*$.

Step 2: Lloyd's algorithm as alternating minimization.

Lloyd's algorithm iterates: 1. **Assignment step**: For fixed centers $\theta^{(t)}$, assign each point to the nearest center. 2. **Update step**: Set $\theta_j^{(t+1)} = \frac{1}{|C_j^{(t)}|}\sum_{i \in C_j^{(t)}} \phi(x_i)$, where $C_j^{(t)}$ is the set of points assigned to center j.

Within the ball B, the assignment step correctly clusters all but an exponentially small fraction of points (by Lemma 3). The update step therefore computes the sample mean of nearly-clean clusters, which is a contraction toward the true cluster means:

$$\|\theta_j^{(t+1)} - \hat{\theta}_{n,j}^*\| \leq \frac{1}{2}\|\theta_j^{(t)} - \hat{\theta}_{n,j}^*\|$$

for all j, with probability at least $1 - \exp(-cn\Delta_{\min}^2/(K\sigma^2 d_\phi))$. (The contraction factor $1/2$ follows from the stability of the Voronoi partition and sub-Gaussian concentration of the sample mean.)

Thus Lloyd's converges linearly to $\hat{\theta}_n^*$ from any initialization in B, achieving accuracy $\Delta_{\min}/16$ in $O(\log n)$ iterations.

Step 3: Random initialization covers the good basin.

Each independent initialization selects K data points uniformly at random (or via k-means++ seeding). The probability that at least one data point from each true cluster is selected is:

$$p_{\text{hit}} = \sum_{\text{permutations } \pi} \prod_{k=1}^{K} \frac{n_{\pi(k)}}{n - \sum_{j<k} n_{\pi(j)}} \geq \prod_{k=1}^{K} \frac{n_k}{n} \geq \pi_{\min}^K > 0$$

where $\pi_{\min} = \min_k \pi_k$ and the inequality uses that selecting a point from each cluster is a coupon-collector-like process. The probability that a selected set of K points includes one from each cluster is at least $\pi_{\min}^K$ (a constant depending only on K and $\pi_{\min}$, not on n).

If the initialization includes a point from each cluster, then using the sub-Gaussian tail bound (Lemma S1), each selected point is within $\Delta_{\min}/8$ of its true cluster center with probability at least $1 - K \cdot \exp(-c\Delta_{\min}^2/\sigma^2)$. Thus a single random initialization lands in B with probability at least $p_0 = \pi_{\min}^K/2$ for large n.

Step 4: Amplification via multiple restarts.

With R independent restarts:

$$P(\text{no restart lands in } B) \leq (1 - p_0)^R$$

Setting $R = C_R \log n$ with $C_R \geq (c+1)/|\log(1-p_0)|$:

$$P(\text{initialization failure}) \leq n^{-c}$$

Each successful initialization converges to within $\Delta_{\min}/16$ of $\hat{\theta}_n^*$ in $O(\log n)$ Lloyd iterations. The total runtime is $O(R \cdot n \cdot K \cdot d_\phi \cdot \log n) = O(n \log^2 n)$, which is polynomial. The solution with the smallest W_n among the R runs is returned. $\square$

6.4 Remarks on Lemma 4

1. **This addresses the NP-hard gap (Issue 3)**: The lemma explicitly proves that under strong separation, Lloyd's with random restarts finds the global empirical minimizer. The proof does not rely on the worst-case NP-hardness of k-means — it exploits the benign landscape induced by well-separated clusters.

2. **If the strong separation condition does not hold**: The result degrades. In the worst case, Lloyd's may find a local minimum that is far from the global minimum. In that case, the theorem would need to be weakened to "there exists a polynomial-time algorithm that finds a c-approximation of the global minimizer," following Kumar & Kannan (2010) or Ostrovsky et al. (2013).

3. **Practical recommendation**: Use k-means++ initialization (Arthur & Vassilvitskii, 2007), which provides $O(\log K)$ approximation in expectation and has better practical coverage than uniform initialization. Lemma 4's conclusion holds for k-means++ as well, with potentially better constants.

4. **The "oracle gap"**: The proof assumes access to the global empirical minimizer in Lemma 2, but Lemma 4 shows that Lloyd's finds it with probability $1 - o(1)$. This closes the gap: the theorem's claim about the global minimizer is also a claim about the Lloyd solution.

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7. Main Theorem: Statement and Proof

7.1 Theorem Statement

Theorem 5 (Fixed-K State Discovery Consistency). Let K be fixed. Suppose:

1. **Generative model**: $\phi(x) = \mu_{s(x)} + \varepsilon$, where $s(x) \in \{1, \dots, K\}$ are the true states and ε is zero-mean sub-Gaussian noise with variance proxy σ^2, independent of $s(x)$.

2. **Strong separation**: $\Delta_{\min} = \min_{i \neq j} \|\mu_i - \mu_j\|_2 > 0$ satisfies

$$\frac{\Delta_{\min}^2}{\sigma^2 d_\phi} \geq C_0$$

   for a sufficiently large universal constant C_0 (the "strong features" regime).

3. **Asymptotics**: $n \to \infty$, and $n_{\min} = \min_k n_k \to \infty$ (every state has infinitely many samples in the limit).

4. **Algorithm**: Lloyd's k-means with $R = C_R \log n$ independent random initializations, returning the solution with the smallest W_n.

Then the estimated partition $\hat{\mathcal{C}}^{(n)}$ satisfies:

$$\begin{aligned} &P(\hat{\mathcal{C}}^{(n)} \neq \mathcal{C}^* \text{ up to permutation}) \\ &\quad \leq K \cdot \exp\left(-c_1 \cdot \frac{n_{\min}\Delta_{\min}^2}{\sigma^2 d_\phi}\right) \; + \; o(1) \end{aligned}$$

where $c_1 > 0$ is a universal constant independent of $K, n, \sigma, \Delta_{\min}$. The $o(1)$ term captures the exponentially small bias from the population minimizer and the initialization failure probability from Lemma 4.

Equivalently, the probability that the estimated partition differs from the true partition (for all points with $\|\varepsilon\| < 3\Delta_{\min}/8$) converges to zero as $n_{\min} \to \infty$.

7.2 Proof of Theorem 5

We combine the four lemmas.

Step 1: Population bias is bounded.

By Lemma 1, the unique population minimizer θ^* satisfies, for some permutation π_1:

$$\|\theta^*_{\pi_1(k)} - \mu_k\| \leq \varepsilon_{\text{pop}} < \frac{\Delta_{\min}}{8}$$

Step 2: Empirical minimizer convergence.

By Lemma 2 (with $t = \Delta_{\min}/8$), for n large enough such that $\Delta_{\min}/8 \geq C_0\sqrt{Kd_\phi/n}$:

$$P\left(\|\hat{\theta}_n - \theta^*\| \geq \frac{\Delta_{\min}}{8}\right) \leq 2 \cdot \exp\left(-c_2 \cdot \frac{n \cdot (\Delta_{\min}/8)^2}{\sigma^2 d_\phi}\right)$$

$$\leq 2 \cdot \exp\left(-\frac{c_2}{64} \cdot \frac{n \cdot \Delta^2_{\min}}{\sigma^2 d_\phi}\right)$$

Combining with Lemma 1 via the triangle inequality:

$$\|\hat{\theta}_n - \mu\| \leq \|\hat{\theta}_n - \theta^*\| + \|\theta^* - \mu\| < \frac{\Delta_{\min}}{8} + \frac{\Delta_{\min}}{8} = \frac{\Delta_{\min}}{4}$$

with probability at least $1 - 2 \cdot \exp\left(-\frac{c_2}{64} \cdot \frac{n \cdot \Delta^2_{\min}}{\sigma^2 d_\phi}\right)$.

Step 3: Lloyd's finds the global minimizer.

By Lemma 4, with $R = C_R \log n$ restarts:

$$P\left(\|\tilde{\theta}_n - \hat{\theta}_n\| \geq \frac{\Delta_{\min}}{16}\right) \leq n^{-c}$$

for any desired $c > 0$. Since $\|\hat{\theta}_n - \mu\| < \Delta_{\min}/4$ with high probability from Step 2, the Lloyd output satisfies:

$$\|\tilde{\theta}_n - \mu\| \leq \|\tilde{\theta}_n - \hat{\theta}_n\| + \|\hat{\theta}_n - \theta^*\| + \|\theta^* - \mu\| < \frac{\Delta_{\min}}{16} + \frac{\Delta_{\min}}{8} + \frac{\Delta_{\min}}{8} = \frac{\Delta_{\min}}{4}$$

with probability at least $1 - 2\exp(-c_2 n \Delta_{\min}^2/(64\sigma^2 d_\phi)) - n^{-c}$.

Step 4: Center proximity implies correct partition.

By Lemma 3, when $\|\tilde{\theta}_j - \mu_{\pi(j)}\| < \Delta_{\min}/4$ for all j, the induced partition matches the true partition for all points with $\|\varepsilon\| < 3\Delta_{\min}/8$.

The fraction of points with $\|\varepsilon\| \geq 3\Delta_{\min}/8$ is bounded by Lemma S1:

$$P(\|\varepsilon\| \geq 3\Delta_{\min}/8) \leq 2\exp\left(-c_0 \cdot \frac{\Delta_{\min}^2}{\sigma^2}\right)$$

This irreducible error is independent of n and is absorbed into the $o(1)$ term. It represents points that are inherently ambiguous due to large noise — even perfect knowledge of the centers cannot classify them correctly.

Step 5: Converting n to $n_{\min}$.

Since K is fixed, $n \geq K \cdot n_{\min}$ (by the pigeonhole principle). Therefore:

$$\exp\left(-\frac{c_2}{64}\cdot\frac{n\cdot\Delta_{\min}^2}{\sigma^2 d_\phi}\right)\leq\exp\left(-\frac{c_2 K}{64}\cdot\frac{n_{\min}\cdot\Delta_{\min}^2}{\sigma^2 d_\phi}\right)$$

Step 6: Final probability bound.

Let $c_1 = c_2K/64$. Collecting all error terms:

$$\begin{aligned}
&P(\text{estimated partition} \neq \text{true partition})\\
&\leq P\left(\|\tilde{\theta}_n-\mu\|\geq\frac{\Delta_{\min}}{4}\right)+P(\text{Lloyd's initialization fails})\\
&\leq 2\cdot\exp\left(-c_2\cdot\frac{n\cdot\Delta_{\min}^2}{64\sigma^2 d_\phi}\right)\ +\ n^{-c}\\
&\leq 2\cdot\exp\left(-c_1\cdot\frac{n_{\min}\cdot\Delta_{\min}^2}{\sigma^2 d_\phi}\right)\ +\ o(1)
\end{aligned}$$

The $o(1)$ term absorbs: - The exponentially small bias ε_{pop} from Lemma 1 (which ensures $\|\theta^*-\mu\| < \Delta_{\min}/8$). - The initialization failure probability n^{-c} from Lemma 4. - The irreducible misclassification from large-noise samples ($\|\varepsilon\|\geq 3\Delta_{\min}/8$), which is $O(\exp(-c\Delta_{\min}^2/\sigma^2))$ and vanishes as $\Delta_{\min}/\sigma\to\infty$. - The K factor in front of the exponential (from the union bound over K centers) is at most $K\leq\exp(c_1 n_{\min}\Delta_{\min}^2/(\sigma^2 d_\phi))^{o(1)}$, so it is absorbed into the exponential.

The final bound:

$$P(\hat{\mathcal{C}}^{(n)}\neq\mathcal{C}^*)\leq K\cdot\exp\left(-c_1\cdot\frac{n_{\min}\Delta_{\min}^2}{\sigma^2 d_\phi}\right)+o(1)$$

This completes the proof of Theorem 5. $\square$

7.3 Verification of Exponents and Inequality Directions

We explicitly verify each critical step:

Step	Inequality	Direction	Verified
Lemma 1, Claim	$P(2\varepsilon^\top(\mu_j - \mu_k) \leq -\|\mu_j - \mu_k\|^2) \leq \exp(-\|\mu_j - \mu_k\|^2/(8\sigma^2))$	$\leq$, correct tail	Yes
Lemma 1, Step 4	$\|\tilde{\theta}_k - \mu_k\| \leq \varepsilon_{\text{pop}}$	$\leq$, upper bound	Yes
Lemma 2, (3)	$\lambda\|\hat{\theta}_n - \theta^*\|^2 \leq \|(P_n - P)(f_{\hat{\theta}_n} - f_{\theta^*})\|$	$\leq$, wrapper around $W_n(\hat{\theta}_n) \leq W_n(\theta^*)$	Yes
Lemma 2, (5)	$\mathbb{E}[\psi_n(r)] \leq C_5\sqrt{Kd_\phi/n} \cdot C_L r$	$\leq$, Dudley	Yes
Lemma 2, (8)	$P(\psi_n(r_j) \geq \lambda r_j^2/4) \leq \exp(-c_6\lambda^2 n r_j^2/(64C_L^2))$	$\leq$, Talagrand	Yes
Lemma 2, sum	$\sum_j \exp(-c \cdot 4^j t^2) \leq 2\exp(-ct^2)$	$\leq$, geometric dominance	Yes
Lemma 3	$\|\phi - \theta_{j_k}\| \leq \Delta_{\min}/2 \leq \|\phi - \theta_{j'}\|$	$\leq$ both sides, correct direction	Yes
Lemma 4	$P(\text{no hit}) \leq (1-p_0)^R \leq n^{-c}$	$\leq$, correct	Yes
Theorem	$P(\hat{\mathcal{C}} \neq \mathcal{C}^*) \leq$ sum of bounds	$\leq$, union bound	Yes

The exponent $-\frac{c_1 n_{\min}\Delta_{\min}^2}{\sigma^2 d_\phi}$ is **always negative** for $n_{\min} > 0$, $\Delta_{\min} > 0$, $\sigma^2 < \infty$, $d_\phi < \infty$. There is no risk of a positive exponent (fixing Review

Issue 2).

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8. Corollary: Sample Size Guide

Corollary (Required samples per state). To achieve misclassification probability $\leq \delta$ due to center estimation, it suffices to have:

$$n_{\min} \geq \frac{C_1 \sigma^2 d_\phi}{\Delta_{\min}^2} \cdot \log\left(\frac{K}{\delta}\right)$$

for a universal constant $C_1 > 0$ (given by the proof constants c_1).

Operational rule of thumb:

Ratio $\Delta_{\min}/(\sigma\sqrt{d_\phi})$	Samples per state for $P(\text{error}) \leq 0.05$
0.5 (marginal)	≥ 400
1.0 (moderate)	≥ 100
2.0 (good)	≥ 25
3.0 (strong)	≥ 12

These numbers assume $K \leq 10$, $d_\phi \leq 64$, and use $C_1 \approx 20$ (from a more detailed constant analysis).

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9. Discussion of Assumptions and Limitations

9.1 Fixed K (Not Growing)

The proof relies on K being fixed. This allows: - The covering number $\log \mathcal{N}(\varepsilon) \leq K d_\phi \log(1 + 2M/\varepsilon)$, which is $O(\log(1/\varepsilon))$ for fixed K. - The strong convexity parameter $\lambda = \pi_{\min}/2$ is fixed (does not decay with K). - The random initialization probability $p_0 \geq \pi_{\min}^K$ is a fixed constant.

For $K \to \infty$ as $n \to \infty$: - The covering number grows like K, giving $K d_\phi$ factors that degrade the bound. - $\pi_{\min}$ could be $O(1/K)$ for balanced states, making $\lambda = O(1/K)$. - The initialization probability decays exponentially in K.

The growing-K regime requires a separate analysis (see Pollard, 1981 for strong consistency with growing K under appropriate conditions; or Lei et al., 2013 for minimax rates).

9.2 The NP-Hard Gap (Issue 3 Resolution)

This proof **does** resolve the NP-hard gap within the strong separation regime:

1. Lemma 2 assumes access to the **global empirical minimizer** $\hat{\theta}_n$.
2. Lemma 4 proves that **Lloyd's algorithm with** $R = O(\log n)$ **random restarts finds** $\hat{\theta}_n$ with probability $1 - o(1)$ under the strong separation condition.

The chain holds because: (a) k-means on well-separated mixtures is not a hard instance — the landscape has a unique local minimum, and Lloyd's contracts toward it; (b) the NP-hardness of k-means applies to worst-case instances, not the well-separated Gaussian-like mixtures considered here.

If the strong separation condition does not hold (i.e., $\Delta_{\min}^2/(\sigma^2 d_\phi)$ is small), then: - The landscape may have spurious local minima. - Lloyd's algorithm may converge to a suboptimal solution. - The theorem's guarantee degrades: one can only claim that there exists some polynomial-time algorithm finding an ε-approximate solution (e.g., via Kumar & Kannan, 2010 or Ostrovsky et al., 2013). - We **acknowledge this gap** and do not claim a guarantee for the weak separation regime.

9.3 Sub-Gaussian Assumption (Issues 9-10)

The proof assumes ε is sub-Gaussian with parameter σ^2. For the bounds: - Lemma S1 uses the sub-Gaussian norm of $\|\varepsilon\|_2$, not the Gaussian chi-squared bound. This fixes Review Issue 10 (the chi-squared bound was only valid for Gaussian vectors). - The required bound $P(\|\varepsilon\| \geq C\sigma(\sqrt{d_\phi} + \sqrt{t})) \leq 2\exp(-t)$ holds for all sub-Gaussian vectors (Vershynin 2018, Theorem 3.1.1). - If ε is only bounded (not sub-Gaussian), the tail bounds degrade from $\exp(-cnt^2)$ to $\exp(-cnt)$ via Hoeffding, changing the exponential rate to a slower exponential. This is a quantitative difference, not a qualitative one — the theorem still holds but with a different constant c_1.

9.4 Known K

The theorem assumes K is known and fixed. In practice, K must be selected (e.g., via the elbow method, gap statistic, or BIC). Estimating K introduces additional uncertainty not captured here. For SCX, K is typically determined by the number of distinct states in the application domain (e.g., phases of a material, regimes of a dynamical system), which is often known from domain knowledge.

9.5 Irreducible Error

Even with perfect center estimates, points with $\|\varepsilon\| \geq 3\Delta_{\min}/8$ can be misclassified. This is bounded by $2\exp(-c_0\Delta_{\min}^2/\sigma^2)$ regardless of sample size. This irreducible error characterizes the **ambiguity of the state partition**: points near decision boundaries are inherently uncertain. The theorem's claim is that the **center estimates converge** and the **partition structure is recovered** — not that every individual point is correctly labeled. The misclassification of boundary points is a feature of the data distribution, not a failure of the estimator.

9.6 Strong Separation Requirement

The condition $\Delta_{\min}^2/(\sigma^2 d_\phi) \geq C_0$ is sufficient but not necessary. It guarantees $\varepsilon_{\text{pop}} < \Delta_{\min}/8$ in Lemma 1. In practice: - If $\Delta_{\min}^2/(\sigma^2 d_\phi)$ is moderate $(1-4)$, the bias ε_{pop} may be larger but could still be $< \Delta_{\min}/8$ for moderate K, d_ϕ. - If $\Delta_{\min}^2/(\sigma^2 d_\phi)$ is small (< 1), the population minimizer may deviate significantly from the true centers, and state discovery may fail. This is the regime covered by Theorem 2 (weak feature failure).

9.7 Proof Constants

The constants c_1, c_2, C_0 in the theorem depend on: - $\pi_{\min}$ (minimum state proportion): enters through $\lambda = \pi_{\min}/2$. - C_L (Lipschitz constant bound): $C_L = 2(M + \sigma\sqrt{d_\phi})$, where $M = \max_k \|\mu_k\|$. - C_5 (Dudley integral constant): a universal constant from empirical process theory. - c_6 (Talagrand concentration constant): a universal constant.

The explicit value of c_1 is:

$$c_1 = \frac{c_6 \pi_{\min}^2 K}{128 \cdot 64 \cdot C_L^2/(\sigma^2 d_\phi)} = O\left(\frac{\pi_{\min}^2 K \sigma^2 d_\phi}{C_L^2}\right)$$

Since $C_L^2 = O(M^2 + \sigma^2 d_\phi)$, the constant depends on the signal-to-noise ratio and cluster balance. For the theorem's purposes, $c_1 > 0$ is sufficient.

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10. References

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15. SCX Theorem 2 (Weak Feature Failure). `theory/theorems/02_weak_feature_failure.md`.

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11. Appendix: Sub-Gaussian Tail Bounds and Quadratic Lower Bound

Lemma S5: Quadratic Lower Bound for W Near θ^*

Statement. Under the conditions of Theorem 5, for any θ with $\|\theta - \theta^*\| \leq \Delta_{\min}/4$:

$$W(\theta) - W(\theta^*) \geq \lambda \cdot \|\theta - \theta^*\|^2$$

where $\lambda = \pi_{\min}/2$ and $\pi_{\min} = \min_k P(S = k)$.

Proof. Let θ satisfy $\|\theta - \theta^*\| \leq \Delta_{\min}/4$. By Lemma 1, $\|\theta^* - \mu\| < \Delta_{\min}/8$, so $\|\theta - \mu\| < \Delta_{\min}/8 + \Delta_{\min}/4 = 3\Delta_{\min}/8 < \Delta_{\min}/2$.

For a point $\phi = \mu_k + \varepsilon$ with $\|\varepsilon\| < \Delta_{\min}/8$, Lemma 3 (applied with $\Delta_{\min}/8$ and $3\Delta_{\min}/8$ bounds) shows that the nearest center in θ is the one matched to true state k. Points with $\|\varepsilon\| \geq \Delta_{\min}/8$ can be assigned differently, but these have probability at most $2\exp(-\Delta_{\min}^2/(128\sigma^2))$ by Lemma S1, which is $o(\pi_{\min})$ under the strong separation condition.

Now consider the region where assignments are correct. For each k, the contribution to W from true state k is:

$$\begin{aligned} &\mathbb{E}[\|\phi - \theta_{\pi(k)}\|^2 \mid S = k] \cdot \pi_k \\ &= \mathbb{E}[\|\mu_k + \varepsilon - \theta_{\pi(k)}\|^2] \cdot \pi_k \\ &= \pi_k \cdot \left(\|\mu_k - \theta_{\pi(k)}\|^2 + \mathbb{E}[\|\varepsilon\|^2]\right) + 2\pi_k \cdot (\mu_k - \theta_{\pi(k)})^\top \mathbb{E}[\varepsilon \mid S = k] \end{aligned}$$

The cross term vanishes since $\mathbb{E}[\varepsilon] = 0$ and $\varepsilon \perp S$. The first term is minimized at $\theta_{\pi(k)} = \mu_k$ with value $\pi_k \cdot \mathbb{E}[\|\varepsilon\|^2]$.

Therefore, in the region of correct assignment (probability $\geq 1-\delta$ for δ exponentially small):

$$W(\theta) - W(\theta^*) \geq \sum_{k=1}^{K} \pi_k \cdot \|\mu_k - \theta_{\pi(k)}\|^2 - \delta \cdot (\text{bounded terms})$$

$$\geq \pi_{\min} \cdot \sum_{k=1}^{K} \|\mu_k - \theta_{\pi(k)}\|^2 - O\left(\exp(-\frac{\Delta_{\min}^2}{128\sigma^2})\right) \cdot (M^2 + \sigma^2 d_\phi)$$

$$\geq \pi_{\min} \cdot \|\theta - \mu\|^2 - \text{negligible}$$

Since $\|\theta^* - \mu\|$ is exponentially small, $\|\theta - \theta^*\| \approx \|\theta - \mu\|$, and:

$$W(\theta) - W(\theta^*) \geq \frac{\pi_{\min}}{2} \cdot \|\theta - \theta^*\|^2$$

for $\|\theta - \theta^*\| \leq \Delta_{\min}/4$ and sufficiently large $\Delta_{\min}^2/(\sigma^2 d_\phi)$. The factor $1/2$ accounts for exponentially small corrections. $\square$

Verification Checklist Against Hostile Review

Review Issue	Status	How v3 Addresses It
Issue 1: Reversed inequality in misclassification bound	**Fixed**	No misclassification bound computed via $O(\sqrt{n/(K \log n)})$ expression. Instead: $P(\text{error}) \leq K \exp(-cn_{\min}\Delta_{\min}^2/(\sigma^2 d_\phi))$ with $\Delta_{\min}$ fixed, giving $P \to 1$. Explicit verification of all inequality directions in Section 7.3.
Issue 2: Positive exponent in uniform convergence	**Fixed**	Replaced covering-number bound with localized empirical process approach (peeling + Talagrand). No polynomial pre-factors in n — only factor 2 from peeling. Exponent $-\frac{c_2 n t^2}{\sigma^2 d_\phi}$ is explicitly negative for all $n, t > 0$.
Issue 3: NP-hard gap of k-means	**Fixed**	Lemma 4 proves Lloyd's with $R = \Omega(\log n)$ random restarts finds the global minimizer with probability $1 - o(1)$ under strong separation. Explicit acknowledgment: without strong separation, the gap remains. Cites Ostrovsky et al. (2013), Kumar & Kannan (2010).

Review Issue	Status	How v3 Addresses It
Issue 4: Lemma 5's incorrect lower bound	**Fixed**	Replaced with self-consistency argument (Lemma 1). No direct computation of $W(\mu) - W(\mu^*)$. Uses sub-Gaussian tail bounds on mis-assignment probabilities (correct inequality direction) and contraction argument.
Issue 5: Fixed vs. scaling $\Delta_{\min}$	**Fixed**	$\Delta_{\min}$ is explicitly fixed (does not scale with n). All scaling analysis removed.
Issue 6c: Triangle inequality noise handling	**Fixed**	Lemma 3 uses $\|\varepsilon\| < 3\Delta_{\min}/8$, giving $\Delta_{\min}/2$ on both sides with correct inequality direction.
Issue 7: Covering number vs VC dimension	**Fixed**	Uses covering number directly (Lemma S3). Acknowledges the bound.
Issue 10: Chi-squared bound for sub-Gaussian	**Fixed**	Lemma S1 uses correct sub-Gaussian norm bound (Vershynin 2018), not Gaussian chi-squared bound.

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